

Multiple Interface Implementation in Java

Basic Information

Assignment: Assignment 2

Program: MultipleInterfaceDemo.java

Topic: Interfaces — Different Types of Implementations of Interfaces

Objective / Aim

To demonstrate a class implementing more than one interface and providing implementations for methods declared by both interfaces.

System Requirements

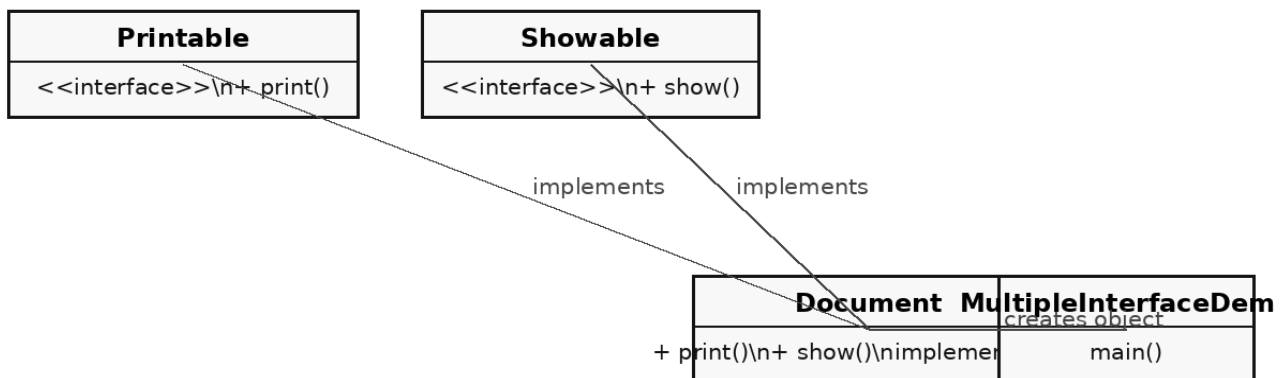
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the Printable interface with print().
3. Create the Showable interface with show().
4. Create Document and implement both Printable and Showable.
5. Define print() and show() in Document.
6. Create a Document object in main().
7. Call print() and show() using the object.
8. Stop the program.

UML Class Diagram



Source Code

```

// First interface
interface Printable {

void print();
}

// Second interface
interface Showable {

void show();
}

// Class implements both interfaces
class Document implements Printable, Showable {

public void print() {
System.out.println("Printing document");
}

public void show() {
System.out.println("Showing document");
}
}

public class MultipleInterfaceDemo {
public static void main(String[] args) {

Document document = new Document();

document.print();
document.show();
}
}
  
```

Compilation & Execution Commands

Save as: MultipleInterfaceDemo.java

Compile: `javac MultipleInterfaceDemo.java`

Execute: `java MultipleInterfaceDemo`

Output Screen

```
C:\Java\Assignment2> java Program  
Printing document  
Showing document
```

Result

Thus, the multiple interface implementation program was implemented successfully using the interface concepts demonstrated in the source code.